

Deep Learning

Autoencoders

Representation Learning

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- ② to learn representations of data
- ③ One way to do so is through Autoencoders

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Example: Synthesizing Human faces

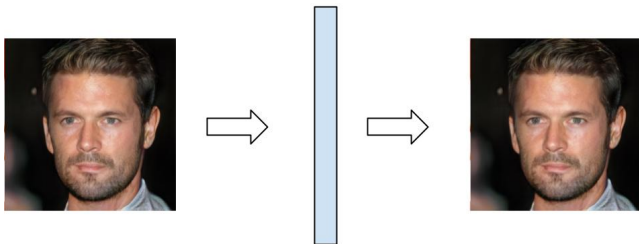
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 - skull size and shape
 - color of skin and eyes
 - features of nose and lips, etc.

Example: Synthesizing Human faces

- ① For generating new faces, it makes sense to capture a small number of degrees of freedom such as
 - skull size and shape
 - color of skin and eyes
 - features of nose and lips, etc.
- ② Even a comprehensive list of such things will be less than the number of pixels in the image (i.e. resolution)

Example: Synthesizing Human faces

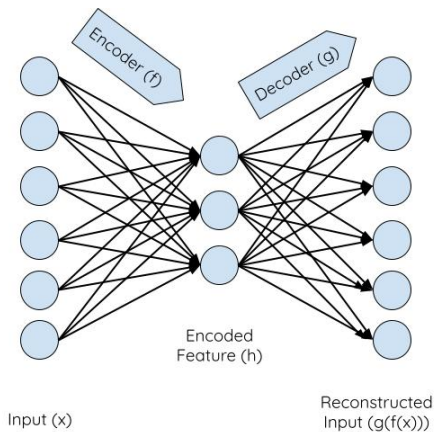
- ① If we can model these relatively small number of dimensions, we can synthesize a face with thousands of dimensions



Autoencoder: architecture

- ① Network consists of two parts: encoder (f) and decoder (g)

Autoencoder: architecture



Autoencoder: principle

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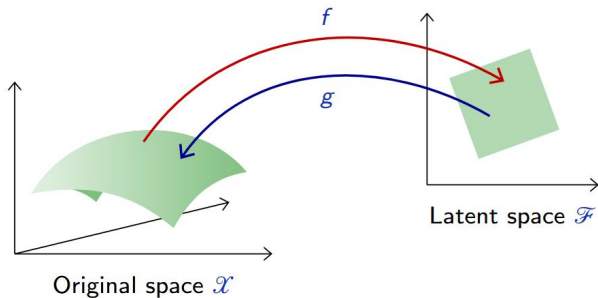


Figure credits: Francois Flucret

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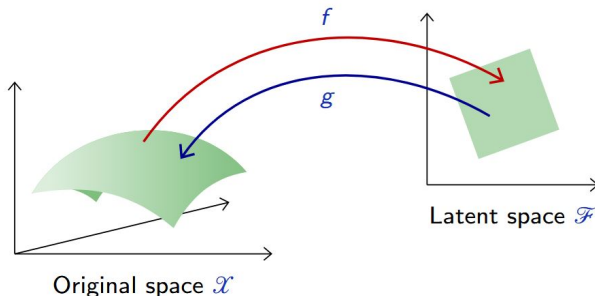


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- ② Loss function takes care of this via training

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$$\mathbb{E}_{x \sim p} \|x - g \circ f(x)\|^2 \approx 0$$

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- ② Training: finding the parameters for the encoder ($f(\cdot; w_f)$) and decoder ($g(\cdot; w_g)$) optimizing the empirical loss

$$\hat{w}_f, \hat{w}_g = \operatorname{argmin}_{w_f, w_g} \frac{1}{N} \sum_n \|x_n - g(f(x_n; w_f); w_g)\|^2$$

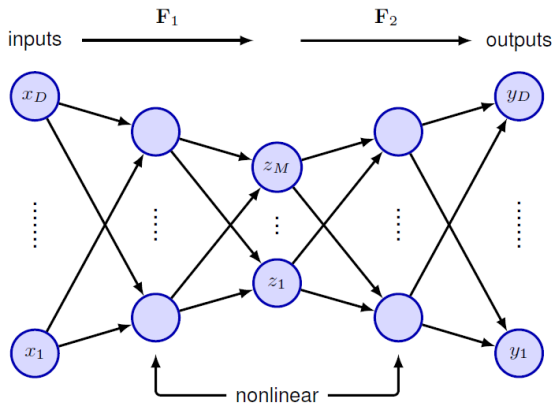
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- ① For binary i/p, we may interpret the reconstructions as probabilities (with a sigmoid nonlinearity)
- ② Hence, we may use BCE loss for training

Deep Autoencoders



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Top row: original data samples

Bottom row: corresponding reconstructed samples (single ReLU layer of dimension 32)

Figure credits: [Keras blog](#)

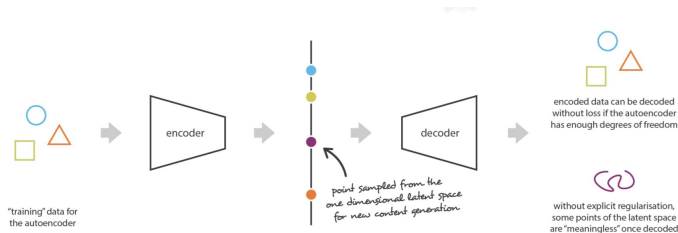
Generative Modeling by Autoencoder

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Generative Modeling by Autoencoder

- ① Reconstructions are not convincing
- ② Because the density model is too simple
 - close points in latent space can give very different decoded data
 - some point of the latent space can give meaningless content once decoded

Generative Modeling by Autoencoder



A good model still needs to capture the empirical distribution on the data, although in a lower dimensional space